Understanding Transformer Architecture

A Technical Guide to Embeddings, Self-Attention, and Feed-Forward Networks

The Transformer architecture is the underlying engine behind almost every modern AI model, including ChatGPT, Claude, and Gemini. Introduced by Google researchers in the seminal 2017 paper *"Attention Is All You Need,"* it replaced older sequential architectures like RNNs and LSTMs by introducing a way to process entire sequences of text simultaneously.

The Core Breakthroughs

- **Parallel Processing:** Instead of reading word-by-word, the Transformer processes every word in a sequence simultaneously. This makes training massive models on GPUs incredibly fast.
- **Self-Attention:** This mechanism allows the model to look at a word and immediately understand its relationship to *every other word* in the sentence, no matter how far away they are.

The Structural Blueprint

The standard Transformer maps an input text sequence through numerical representations and structural blocks to build context-aware representations:

```
[Input Text] → [Positional Encoding] → [Encoder Blocks] → [Decoder Blocks] → [Output Text]
```

1. Vector Embedding (The Raw Coordinates)

Before an AI can read words, it must translate them into numbers. A **Vector Embedding** takes a word and plots it as a point in a massive, multi-dimensional mathematical landscape. Words with similar definitions are placed close together in this space. However, at this initial stage, the embedding is static—the model cannot distinguish context (e.g., whether "bank" means a financial institution or a riverbank).

Word Embeddings Example Matrix Look:

```
"River" → [ 0.74, -0.12, 0.89, ... ] (Near "water", "stream")
"Bank" → [ 0.11, 0.45, -0.23, ... ] (Ambiguous mid-point)
```

2. Self-Attention Mechanism (The Context Engine)

Self-Attention dynamically updates each word's embedding by looking at the words surrounding it. For every word, the system generates three vectors: **Query (Q)** (What am I looking for?), **Key (K)** (What information do I contain?), and **Value (V)** (What substance do I offer?).

- **The Query Check:** The query vector of a target word is compared mathematically (dot-product) against the key vectors of all words in the sequence.
- **Attention Scores:** The results are scaled and passed through a Softmax function to turn them into dynamic percentages totaling 100%.
- **The Final Mix:** The model multiplies these percentages by the Value vectors of each word and adds them together. The vector for "bank" explicitly absorbs the mathematical essence of "river", updating its location in the vector space.

3. Feed-Forward Neural Network (The Fact-Checker)

Once Self-Attention finishes blending context across tokens, the updated vectors pass into the **Feed-Forward Neural Network (FFNN)**. While Self-Attention lets words interact, the FFNN treats each position completely independently. It applies two linear transformations with a non-linear activation layer (like ReLU or GELU) in between, allowing the model to synthesize the gathered context against its internal world knowledge database.

Architectural Architecture Evolution Summary

Feature	Old Architectures (RNN/LSTM)	Transformer Architecture
Processing Style	Sequential (one word at a time)	Parallel (all words at once)
Training Speed	Slow (cannot be easily distributed)	Extremely Fast (highly optimized for GPUs)
Context Handling	Struggles with long-range dependencies	Flawlessly links words across massive windows